



From housing gains to pension losses: Micro-macro integration to reveal wealth inequality dynamics in Chile

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ABSTRACT

This paper examines the levels and dynamics of wealth inequality in Chile between 2007 and 2021, in a context where the private pension system is mandatory and government housing policies are centered on private ownership. We focus on two major macroeconomic events: the sharp increase in housing prices following the introduction of a value-added tax on new dwellings in 2016 and the large-scale liquidation of pension assets through early withdrawals during the recent pandemic. To do so, we construct a micro–macro consistent wealth distribution by extending existing methodologies to integrate administrative pension fund balances into household wealth surveys using machine learning techniques. Our results reveal pronounced wealth concentration: the top 1% hold roughly one-third of total private wealth—levels comparable to those observed in the United States—driven by their disproportionate ownership of financial assets. By contrast, the bottom 50% hold 7–9%, primarily through housing and pension assets, shares that exceed those reported in most European countries. We also document a modest decline in wealth inequality over the period, particularly after 2016. A counterfactual analysis suggests that this decline reflects the interaction of two opposing forces: housing appreciation, which predominantly benefited the bottom 90% of households, and pension withdrawals, which reduced wealth across the distribution while relatively increasing inequality. Our findings on levels, trends, and counterfactuals are robust across alternative methodological specifications.

1. Introduction

The recent wealth literature has expanded our empirical understanding of inequality by integrating household surveys, tax records, and national accounts into unified frameworks (Piketty, 2014; Zucman, 2019; Smith et al., 2023). This work has highlighted the central role of asset composition—particularly financial assets, housing and pension wealth—in shaping long-run distributional trends (Saez and Zucman, 2016; Martínez-Toledano, 2023; Martínez-Toledano et al., 2023). Despite these advances, empirical evidence remains heavily concentrated in a small set of countries with sufficiently developed statistical systems, where detailed tax microdata, comprehensive national accounts, and high-quality household surveys allow accurate measurement of assets and liabilities. As a result, much of the developing world remains uncharted territory in the study of wealth inequality, limiting our ability to generalize about the processes of wealth accumulation under different institutional settings.

To expand on the comprehension of wealth inequality dynamics in the global south, we present the first micro–macro consistent estimates of wealth inequality in Chile for 2007–2021. These estimates capture a period of accelerated wealth growth shaped by two rare policy-induced shocks. The first was the surge in housing prices partly triggered by the introduction of a value-added tax (VAT) on new dwellings in 2016 (Lozano and Idrovo, 2024). The second was the partial liquidation of pension assets following the early withdrawals authorized during the COVID-19 pandemic. We analyze the distributional consequences of both episodes in a context defined by a mandatory, privately-managed pension system and extensive ownership-based housing subsidies, two enduring legacies of the privatization reforms introduced under the Pinochet dictatorship.

To construct our empirical contribution, we combine data from the national household wealth survey (*Encuesta Financiera de Hogares*, EFH), a representative sample of administrative pension-fund records, financial balance sheets from national accounts, and Forbes billionaire

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Fig. 1. The evolution of Chilean private net wealth (2007–2021). **Notes.** Panels A and B: Elaborated by the authors using data from financial accounts (Central Bank), the housing price index (Central Bank and the National Tax Agency), and Flores et al. (2018). Panel A: The market value of total wealth, net of liabilities, is approximately four times the National Income in 2021. The “Real Estate” category includes formal housing and agricultural land, and the “Financial Assets” category comprises stocks, cash, and other financial assets. Panel B: All categories are expressed in 2021 billions of nominal USD and are normalized such that the value for 2007 equals 1. Panels C and D: Constructed by the authors using survey data, imputed pension funds and cash, and the top-correction approach proposed by Vermeulen (2016).

lists. To complement the survey, which does not consistently capture pension balances despite their central role in household wealth, we integrate an administrative sample of individuals enrolled in the Chilean pension system. We use these administrative records to train, validate and contrast several families of machine-learning models to impute pension wealth for survey respondents, complementing standard top-correction and uprating techniques to align the survey with national wealth totals.¹ To fill the absence of non-financial assets in official national balance sheets, we construct new macroeconomic estimates of real estate, i.e., housing wealth and agricultural land. To benchmark our estimated wealth inequality levels, we use individual administrative data from the Chilean tax authority, which provide a detailed snapshot of the upper tail—including direct and indirect firm ownership—for a single year.² In this regard, our main contribution lies in constructing the

levels and dynamics of wealth inequality in Chile, the careful treatment of private pension funds, and in documenting their effect on the wealth distribution.

As a preview for our distributional analysis, Fig. 1 shows the evolution of national private wealth, net of liabilities, between 2007 and 2021, together with our preferred measures of wealth inequality. Panel (a) illustrates a marked increase in the wealth-to-national-income ratio, which rose from about 3.2 to just above 4 over the fourteen-year period. Real estate constitutes the largest component, accounting for roughly half of total private wealth, while pension funds and financial assets each represent about a quarter. Panel (b) presents the growth of each asset category as a normalized index. During the first half of the period, large asset classes expanded at comparable rates, with rising liabilities offsetting much of this growth and keeping aggregate net wealth relatively stable. After 2016—coinciding with the introduction of the 15 percent VAT on new housing—net wealth accelerated, mainly driven by real estate. Between 2019 and 2021, however, total wealth to income ratio declined by 3.7 percent, largely due to a contraction in pension assets, which fell by roughly 5 percent in 2020 and a further 18 percent in 2021. This drop reflects the extraordinary pension withdrawals authorized universally as part of the emergency relief measures during the COVID-19 pandemic. These macroeconomic shifts in real estate and pension wealth are expected to be key determinants of Chile’s recent wealth inequality dynamics.

Panels (c) and (d) show our baseline distributional levels and trends estimated for Chile for the top 1 percent and the bottom 50 percent. We show that the wealth share of the top 1% in Chile ranges between 31.6% and 38%, placing the country close to the United States in

¹ Our method is inspired by the two-sample, two-stage least squares (TS2SLS) approach commonly used to study intergenerational mobility under data scarcity (Kenedi and Sirugue, 2023; Jácome et al., 2025). In its original form, the method links independent samples of parents and children by exploiting common covariates such as education, occupation, or region. More recent applications combine this framework with machine learning techniques to improve predictive accuracy—for example, Bloise et al. (2021) use it to impute parental earnings in South Africa, while Britto et al. (2025) apply it to predict informal income in Brazil.

² Access to these data was granted as part of a tax reform study conducted by the Chilean Finance Ministry under the supervision of Damián Vergara. The data were used to compute the denominator of top wealth shares, complemented by macroeconomic aggregates presented later in the paper.

terms of wealth concentration (Saez and Zucman, 2016; Smith et al., 2023). Excluding mandatory pension assets from the wealth definition increases the top 1% share by 7.6 to 13 percentage points, indicating that mandatory pensions play a substantial equalizing role in the wealth distribution. By contrast, the bottom 50% of households hold between 7% and 9% of total wealth.

Regarding wealth inequality trends, we document a slight decline in wealth inequality between 2007 and 2021. Two opposing forces appear to drive these dynamics: one that increased asset values and another that led to their liquidation. To explore these mechanisms, we conduct two counterfactual analyses following the DFL reweighting approach proposed by DiNardo et al. (1996). In the first counterfactual, we assume that housing asset growth after 2016 followed the same trend observed prior to that year. In the second scenario, we assume that pension levels and trends during the pandemic remained consistent with their pre-2020 trajectory. The results indicate that, absent the post-2016 rise in housing wealth, the decline in inequality between 2014 and 2021 would not have occurred. Conversely, in the absence of pension wealth reductions, the equalizing effect driven by rising housing wealth would have been even stronger.

These findings suggest that the increase in housing prices—while widespread—was relatively faster toward the bottom of the wealth distribution, generating an equalizing effect. By contrast, the distributional consequences of pension fund withdrawals are less straightforward. Following the withdrawals, pension assets declined across the entire distribution. Because pension wealth is the most evenly distributed component of household wealth, its contraction mechanically increased overall inequality by increasing the relative importance of more concentrated assets, such as housing and financial wealth, in household portfolios. Taken together, our results imply that government policies aimed at raising revenues or smoothing consumption in response to major economic shocks can have indirect but quantitatively meaningful effects on the wealth distribution.

When comparing inequality estimates across countries, we find that the bottom 50% of households hold a considerably larger share than that observed in France and Germany (Garbinti et al., 2021; Albers et al., 2024) and closer to the levels found in Spain (Martínez-Toledano, 2023). This difference reflects the institutional structure of the Chilean economic model, which emphasizes ownership and compels households to accumulate wealth primarily through housing and mandatory pension savings, whereas European systems rely more heavily on in-kind transfers and unfunded public pensions.³ This contrast is also evident in portfolio composition. In Germany (Albers et al., 2024), the bottom 50% of households hold most of their wealth in bank deposits, while in France (Garbinti et al., 2021) and Spain (Martínez-Toledano, 2023) it is concentrated in primary housing. In Chile, the bottom half of the distribution holds pension and housing wealth in roughly equal proportions, with lower cash savings.

Despite the equalizing forces associated with ownership-based social security, wealth concentration at the top remains high. This is driven by the extreme concentration of financial assets among the wealthiest households, consistent with the high levels of capital income concentration documented in Chile (Flores et al., 2020; De Rosa et al., 2024). These findings suggest that ownership-oriented social security policies, while effective at raising individual wealth holdings among the lower half of the distribution, are insufficient on their own to mitigate extreme wealth concentration at the top.

³ Social protection can be structured either through publicly provided services or through market-oriented policies that subsidize private ownership. In housing, for example, governments may offer rental subsidies and social housing or, alternatively, ownership subsidies; in pensions, the contrast lies between pay-as-you-go schemes and fully funded individual accounts. These institutional choices have distributional consequences: public provision tends to concentrate resources in the public sector, whereas market-oriented models translate support into privately held assets.

Our results are robust to a set of different specifications in the construction of both aggregate and distributive series. Specifically, we adjust the valuation of unlisted firms by combining price-to-earnings ratios from the Chilean stock exchange with aggregate tax data on corporate earnings, following Albers et al. (2024). We also incorporate an experimental series on household wealth from the Central Bank of Chile (Balsa and Vásquez, 2023) as an independent benchmark. Beyond these aggregate adjustments, we implement multiple calibration procedures to align the household survey with national wealth totals, test alternative machine-learning specifications for imputing pension balances, and conduct sensitivity analyses on the parameters used in the top-correction procedure. Collectively, these methodological exercises yield robust and internally consistent estimates of wealth inequality levels and trends in data-constrained contexts, particularly in settings where private pension assets represent a major component of household wealth.

Related Literature

We contribute to several strands of the wealth inequality literature. First, we extend and complement research on wealth-inequality measurement (Saez and Zucman, 2016, 2020; Kopczuk and Saez, 2004; Kopczuk, 2015; Smith et al., 2023; Jakobsen et al., 2020; Acciari et al., 2024; Garbinti et al., 2021; Alvaredo et al., 2018; Martínez-Toledano, 2023; De Rosa, 2025; Albers et al., 2024; Advani et al., 2025; Hempel, 2022) by providing new evidence on micro-macro-consistent wealth-distribution dynamics in a developing-economy context. Specifically, we document 14 years of wealth-inequality trends in Chile and show how these patterns evolve under a social security system that relies heavily on private ownership of pension savings and housing assets. Moreover, we trace changes in the distribution of wealth to major policy shifts affecting housing (through VAT reforms) and pensions (through emergency withdrawals), highlighting their differential impacts across the wealth distribution.

Our findings on wealth inequality trends and portfolio composition in Chile can be understood within a broader international context. Indeed, in the United States, the share of wealth held by the top 1% rose markedly from the mid-1980s until the eve of the COVID-19 pandemic (Smith et al., 2023; Saez and Zucman, 2016). This increase has been attributed primarily to the growing importance of fixed-income and business assets in total wealth. However, recent evidence shows a decline in wealth inequality during the pandemic (Kuhn and Ríos-Rull, 2025), largely explained by rising housing prices that disproportionately benefited households in the bottom 90% of the wealth distribution. Historical U.S. data also highlight the central role of housing and pensions in reducing wealth inequality between the 1930s and the mid-1980s (Saez and Zucman, 2016)—a pattern that closely parallels our findings for Chile.

The relationship between housing cycles and wealth inequality has also been documented in Europe. In Spain, Martínez-Toledano (2023) shows that housing booms reduce wealth inequality, while housing busts increase it. For example, between 2007 and 2014, the share of the top 10% increased while that of the bottom 90% declined. Similarly, during Spain's housing boom of the late 1990s, the national wealth-to-income ratio rose from 4 to 6. A comparable dynamic occurred in France and Germany (Garbinti et al., 2021; Albers et al., 2024): in France, the ratio increased from 3.2 to 5.5 between 2000 and 2014, largely driven by housing prices, and in Germany, it rose from 4 to 7 between 2010 and 2018, fueled by both housing and business assets. Our results also extend and complement those of Martínez-Toledano et al. (2023), who show that in the Netherlands, private pensions play a strongly equalizing role in the income distribution, particularly among the bottom 50%, a pattern we also observe for Chile.⁴

⁴ Recent evidence also documents a decline in the wealth share of the bottom 50% in Italy (Acciari et al., 2024), a reduction in top-wealth inequality in the United Kingdom (Advani et al., 2025), and a slight decrease in Canada (Hempel, 2022). Notably, the latter two countries are characterized by pension systems that play a substantial role in household wealth accumulation.

Only a few studies on wealth inequality have focused on developing countries, largely due to severe data limitations. For instance, Carranza et al. (2025) highlight the scarcity, incompleteness, and inconsistencies of reliable wealth data across Latin America. Where measurement is possible, however, concentration levels appear extreme: the top 1% hold between 37% and 40.6% of national private wealth, placing the region among the most unequal worldwide. Similarly, based solely on survey data, Gandelman and Lluberá (2024) examine the relationship between wealth indicators and the sociodemographic characteristics of household heads in countries such as Chile, Colombia, Mexico, and Uruguay. De Rosa (2025) estimates aggregate wealth and its distribution in Uruguay, Chatterjee et al. (2020) conduct a similar exercise for South Africa, and Piketty et al. (2019) provide comparable evidence for China.

Our work builds on Flores and Gutiérrez (2021), who previously estimated an aggregate wealth series for Chile. We extend and complement this contribution by constructing a longer and more comprehensive wealth series that incorporates durable wealth stocks and by producing alternative estimates correcting for unlisted business assets. In addition, Carranza et al. (2023) estimate the share of the top 1% using data from the Chilean Ministry of Finance's tax reform study combined with the wealth series from Flores and Gutiérrez (2021). Within this context, our study contributes to this literature by constructing, for the first time, a micro–macro consistent wealth distribution for Chile, enabling the estimation of wealth inequality levels and trends that are robust to multiple methodological decisions.

In addition, we contribute to the literature on imputation methods in wealth inequality research (Vermeulen, 2016; Cowell and Kerm, 2016; Alvaredo et al., 2018; Saez and Zucman, 2016) by implementing a two-stage approach, similar to Kenedi and Sirugue (2023) and Jácome et al. (2025) used in intergenerational mobility literature, that leverages administrative pension data to predict private pension balances within the household survey. This procedure applies machine learning techniques in line with Bloise et al. (2021) and Britto et al. (2025), enabling the imputation of pension wealth at the individual level.

Recent advances in the inequality literature increasingly rely on machine learning (ML) methods to impute wealth and income. For instance, Suss et al. (2023) use ML algorithms to develop accurate predictive models of household wealth, while Brunori et al. (2023) compare alternative approaches to addressing missing observations and show that sample reweighting based on non-response probabilities is particularly effective. Similarly, Lucchetti et al. (2024) introduce a Least Absolute Shrinkage and Selection Operator with multiple imputation by Predictive Mean Matching (LASSO–PMM) to estimate intra-generational income dynamics, and Kukuk et al. (2023) compare ML models to estimate gender wealth gaps across Europe. In this regard, the methods developed in this paper can be directly applied to improve wealth inequality estimates in countries with mandatory private pension systems, such as the Netherlands and Peru.

The paper is organized as follows: Section 2 provides details about the national context and our main data sources. Section 3 describes the methodology used to derive our results. The main results are presented in Section 4. Section 5 provides an explanation for our results by presenting counterfactuals. Finally, Section 6 synthesizes the results and provides a brief conclusion.

2. Data, concepts, and context

This section presents the Chilean institutional context for private pensions and social housing. It also presents wealth aggregates and the household wealth survey, its limitations, and the data used to complement it.

2.1. The Chilean institutional context

Chile's institutional framework reflects a distinctive conception of the State, shaped during the military dictatorship of Augusto Pinochet (1973–1990), when a subsidiary model replaced the collective and

redistributive principles that had guided much of the 20th century. This for-profit, market-oriented architecture gives ground to the organization of current social policy and public services in the country, even though they were later adjusted but not overturned by democratic governments. The following subsections describe the current pension and housing systems, two central pillars of this institutional architecture and two main asset categories shaping the country's macroeconomic and distributional landscape.

2.1.1. The Chilean pension system

The current pension system in Chile was created in 1980, in the middle of the military dictatorship, and entered into force one year later. The system is composed of a mandatory private component based on individual capitalization and, more recently, a government-subsidized component.⁵ Under the private system, individuals are required to contribute 10 percent of their pre-tax monthly wages to an individual pension account and may freely choose both their pension fund administrator and their preferred level of investment risk, with a limited choice of five regulated composite funds. The system caps the pre-tax income subject to mandatory contributions at approximately USD 3000 per month. Beyond this threshold, pension assets represent a diminishing share of household portfolios, as higher-income individuals are not required to contribute on earnings above the cap. Although voluntary contributions are allowed, and are tax-free until withdrawal, they remain limited and represent less than 10% of total contributions (see Appendix D.1.5 and Troncoso, 2022).⁶ Prior to 2019, participation was voluntary for self-employed workers. Despite an informality rate of approximately 30 percent, informality spells are short –3.5 months on average– so we should expect most working-age individuals to hold pension accounts (Villena-Roldán, 2024). Mandatory pension savings are managed by *Administradoras de Fondos de Pensiones* (AFPs), which are for-profit entities responsible for investing contributors' funds in financial assets.

The system sets eligibility for retirement at 65 years for men and 60 years for women. Retirees can choose between annuities and programmed withdrawals. In an annuity, the retiree transfers their savings to an insurance company, which guarantees a lifelong monthly payment, assuming all longevity risks. In programmed withdrawals, the retiree keeps their funds in their pension account and receives a monthly pension recalculated annually based on the remaining balance, expected lifetime, and market returns. The pension may decrease over time, but any funds left upon death become part of the retiree's property, thus inheritable.

When an account holder dies, the pension system transfers accumulated funds through survivor pensions or inheritance. By law, survivor pensions are granted to spouses or civil partners; children under age 18 (or under age 24 if enrolled in full-time education); children with disabilities, regardless of age; and financially dependent parents. If no eligible survivors exist, or if residual balances remain after survivor pensions are paid, legal heirs may withdraw the remaining funds as a lump-sum payment.⁷

⁵ Police and military personnel are covered by different regimes, which are more generous and publicly funded. Together, these groups account for only about 1.2% of the adult population.

⁶ In addition to mandatory contributions, Chile's pension system allows workers to make voluntary pension savings through the *Ahorro Previsional Voluntario* (APV) scheme, which offers two alternative tax regimes. Under Regime A, the government directly complements voluntary savings with a 15 percent bonus on the contributed amount, subject to an annual cap. In contrast, Regime B provides an income tax incentive, allowing savers to deduct voluntary pension contributions from their taxable income up to another cap.

⁷ An exception allows for early withdrawal of pension funds in cases of terminal illness. When there is medical certification indicating a high probability of death within 12 months, individuals are permitted to access their pension savings prior to retirement.

Over time, the system has evolved into a mixed model with an important subsidiary component. In 2008, the individual capitalization pillar was complemented by a public, means-tested component, with further improvements in more recent years, outside our period.⁸

Pension withdrawals reform

In response to the COVID-19 pandemic, the Chilean Congress approved a first reform authorizing partial withdrawals from individual pension accounts in July 2020, followed by two additional rounds in December 2020 and May 2021. Collectively, these three withdrawals amounted to nearly 19% of Chile's GDP.⁹ Each round allowed individuals to withdraw up to 10% of their pension balance, subject to upper and lower limits. Specifically, when 10% of an individual's balance fell below approximately USD 1,000, the full amount could be withdrawn; conversely, when it exceeded roughly USD 4,300, withdrawals were capped at that value. As a result, individuals with balances below USD 10,000 could liquidate a relatively larger share of their pension savings, while those with balances above USD 43,000 were restricted to smaller proportional withdrawals. Approximately 10 million individuals participated in the first withdrawal round.

This figure exceeds the total active labor force reported by the World Bank for the same year (9.17 million), implying that withdrawals were made not only by virtually all active contributors but also by pensioners and other inactive individuals.¹⁰ Participation remained high in the subsequent rounds, though it declined slightly—from approximately 9 million participants in the second round to about 8 million in the third.

Private pension funds as household wealth

Although private pensions are mandatory, illiquid and partly outside households' control, they are central to private wealth, especially in the case of Chile. Balances carry insurance and bequest value because they can be inherited, which parallels the standard treatment of life insurance in the wealth-inequality literature (Saez and Zucman, 2016; Smith et al., 2023; Martínez-Toledano, 2023). Moreover, pensions can become liquid under policy discretion, as seen during COVID and in Perú's 2025 authorization of an eighth withdrawal (Congreso de la República del Perú, 2025). Conceptually, mandatory pension saving substitutes for voluntary saving, so the expected present value of pension claims influences consumption and saving choices (Martínez-Toledano et al., 2023). Empirically, including pensions can materially change measured inequality depending on the relative size of pension funds, as shown for the Netherlands and Canada (Martínez-Toledano et al., 2023; Hempel, 2022). In line with the standard approach in the literature, we include private pensions in the definition of private wealth, while stressing that their measurement—often treated as a second order issue—requires particular care in institutional settings like Chile's.

2.1.2. Housing policy and subsidies in Chile

The Chilean housing subsidy system stands in sharp contrast to European social housing systems, where states typically support low-income households through transfers and in-kind provision (Bérard and Trannoy, 2025). The Chilean model is strongly oriented toward ownership through demand-side subsidies. These structural differences in

subsidy design generate significant differences in wealth distribution, particularly at the lower end of the distribution.

The military dictatorship introduced demand-side housing vouchers in 1977, replacing the state-led model with market-oriented housing policy.¹¹ This allowed eligible households to combine subsidies with savings and mortgage credit to purchase housing. After 1990, successive democratic governments expanded and diversified the voucher subsidy system. A central initiative was the *Programa de Vivienda Dinámica sin Deuda* (PVD), which targeted the poorest households excluded from mortgage markets.¹² In the 2000s and 2010s, reforms increasingly aimed at middle-income households. The Middle-Class subsidy (*Subsidio para la Clase Media*) extended state support to households that did not qualify as poor but lacked sufficient credit access, thereby promoting broader homeownership.¹³

Home-ownership is high throughout the income distribution, both in urban and rural areas (Table C.3). Rates are on the order of 60–70% even in the bottom half, and they remain of a similar magnitude at the top. Across income groups, levels typically differ by no more than 10–15 percentage points. Rural home-ownership is higher than urban home-ownership at every point of the income distribution.

VAT reform on housing

Chile's housing value-added tax (VAT) was first introduced in 1987. From the beginning it encompassed the sale of new dwellings built and sold directly by construction firms. This rule remained in place for nearly three decades, yet it only affected a narrow segment of the market. The 2014 tax reform (effective in 2016) broadened the scope of the tax to include transactions by habitual sellers—that is, companies or individuals who regularly buy and sell properties as a business, such as those selling within a year of acquisition or construction. The reform also curtailed the long-standing Special Tax Credit for Construction Companies, a measure that had allowed builders to offset part of the sales-tax burden to encourage housing supply. After 2016, the credit was limited to dwellings valued below roughly USD 74,000 (as of 2017) and capped at USD 8350 per unit. The VAT was also extended to lease-to-own contracts, while exemptions remained for homes with building permits issued before January 1, 2016, government-subsidized housing, and property transfers through inheritance or donation.

As Lozano and Idrovo (2024) document, the reform increased the prices of new dwellings by 9.6–12.6%, implying a substantial pass-through of the 19% VAT to housing prices. Although their analysis focuses on a specific segment of the housing market, price increases are also likely to have affected transactions of second-hand dwellings through market-wide spillovers. While we do not attribute the entire observed increase in real estate to the reform—other factors, such as the recent expansion of public transport to peripheral urban areas, particularly the Santiago subway system, may also have contributed to rising prices—we view the VAT reform as a major driver of the housing wealth dynamics that play a key role in our empirical results.

⁸ In 2022, the government expanded this scheme through the *Pensión Garantizada Universal* (PGU), a monthly benefit aimed at preventing old-age poverty. The program covers individuals belonging to the bottom 90% of the household income distribution.

⁹ These measures prompted large-scale liquidity operations by the Central Bank of Chile, contributed to tighter labor market conditions, and are expected to reduce contributory pensions by an average of 21% for future retirees (Briones et al., 2023; Inzunza and Madeira, 2023).

¹⁰ See https://www.spensiones.cl/portal/institucional/594/articles-14743_recurso_1.pdf.

¹¹ Chile's housing policy dates to the early 20th century *Cuestión Social*, when rapid urbanization led policymakers to view housing shortages as sources of public health risks and social unrest. The state responded by creating the *Caja de la Habitación Popular* (1936) and later the *Corporación de la Vivienda* (CORVI), which led large-scale public housing programs in the 1960s under the governments of Eduardo Frei Montalva and Salvador Allende; see Appendix E.

¹² Under this program, families received subsidies that fully financed basic housing units without debt. Units were modest in size (30–40 square meters) but designed for progressive expansion.

¹³ The Social and Territorial Integration Subsidy (DS19), introduced in 2016, finances the purchase of new dwellings in socially integrated projects for non-homeowners registered in the Social Registry of Households by providing a direct subsidy that reduces the amount to be financed. Following the 2010 earthquake, additional rental and reconstruction subsidies were introduced.

2.2. Wealth aggregates: concepts and definitions

We measure *private wealth* as the net wealth of households and non-profit institutions serving households (NPISH) valued at market prices.¹⁴ Net private wealth W_t is defined as all assets minus liabilities L_t . We distinguish three asset blocks: non-financial assets A_t^{NF} , financial assets excluding pensions A_t^F , and pension wealth P_t . Total private wealth in year t is

$$W_t = A_t^{NF} + A_t^F + P_t - L_t \quad ; \quad A_t^{NF} = H_t + F_t + D_t \quad (1)$$

Non-financial (or “real”) assets can be decomposed into real estate and durable goods. Real estate includes housing assets H_t —the value of residential dwellings and associated land—and agricultural land F_t . Durable goods D_t include vehicles, farm assets, and machinery and equipment used by self-employed workers and family enterprises. These items often represent tangible capital held in unincorporated businesses and therefore overlap with household balance sheets.

Because no single official housing series covers the full period, we construct residential property wealth H_t using a composite of administrative sources anchored to the benchmark estimates in Flores et al. (2018). Their study is an ideal reference point, as it combines the full national cadastre with administrative records covering the universe of effective housing transactions in each year, and applies a hedonic-pricing approach to recover market-consistent values. We take their series as our baseline, extrapolating backward to cover years prior to 2012 and forward to cover the period after 2017 using administrative data on construction permits, which report average dwelling size and the evolution of market prices per square meter. Appendices A.1 and D.1 elaborate on the methods and data employed to estimate housing wealth, including robustness checks on the presence of informal housing. They also present a comparison between our estimates and experimental housing stock statistics recently published by the Chilean Central Bank.

To estimate the market value of agricultural land F_t , we start from the volume reported by the Internal Revenue Service in aggregate territorial tax reports. While the administrative counts are reliable, fiscal valuations are expected to be substantially below market values. We therefore assume that the proportional undervaluation of agricultural land is similar to that observed for housing. Concretely, we compute an adjustment factor as the ratio of the market value of housing (above) to the total fiscal value of dwellings reported, and apply this factor to the fiscal value of agricultural land. This yields a correction multiplier between 1.5 and 2.4 depending on the year. The multiplier declines over time, reflecting reforms to fiscal assessment rules that narrowed the gap with market values.

Our benchmark for financial assets excluding pensions A_t^F is the Central Bank of Chile’s Financial Accounts, compiled under the 2008 SNA guidelines, which provide a common framework with most estimates in the literature. They include currency and deposits, debt securities, equity and investment fund shares (listed and unlisted), insurance technical reserves excluding pensions, and other receivables (e.g., loans granted and other accounts receivable).¹⁵ We treat pension wealth separately from A_t^F because it differs in liquidity, institutional design,

and in how it can be reconciled with household survey microdata, as P_t is expected to track the labor income distribution more closely than other financial assets.

Our reference for pension wealth P_t also comes from official Financial Accounts, which are compiled directly from administrative records from the Superintendency of Pensions on the actual individual funds with direct observations of their market prices. Appendix A.3 provides more details on the construction of financial accounts, their underlying sources, and our alternative estimates, including both pension and non-pension wealth and its subcomponents.

Liabilities (or Debt) are measured using the Financial Accounts and correspond to household loans. They consist primarily of mortgages and consumer credit. Loans are classified by original maturity into short-term loans, defined as those with an original maturity of one year or less and long-term loans, defined as those with an original maturity exceeding one year.

We construct durable goods D_t using the Household Finance Survey. Because our top-correction procedure introduces additional synthetic units, it may mechanically affect aggregate D_t . We therefore report aggregates both including and excluding the durable wealth generated by the top-correction methodology. Detailed results are presented in Appendix A.4.

We distinguish between categories that are fully reconcilable with national accounts and supplementary items that capture additional but less standardized wealth components. In our setting, the Financial Accounts provide the integrated benchmarks for A_t^F , P_t , and L_t . Our estimates of H_t and F_t should also, in principle, align with the System of National Accounts. However, the informal component of wealth we use as a robustness check and D_t , extend the standard accounting perimeter. The estimation of illegal, offshore, or unreported holdings remains an important topic, yet we acknowledge this as a limitation of our study, as we lack reliable data to estimate the level and distribution of such assets.

2.3. Distributional wealth data

2.3.1. The household wealth survey

The Household Finance Survey conducted by the Central Bank of Chile provides detailed household balance sheets for years 2007, 2011, 2014, 2017, and 2021. It is representative of the urban population, which accounted for 87.8% of the population in the 2017 Census.

Despite its strengths, the survey has some limitations that require supplementation. First, coverage of the top of the wealth distribution is limited. Although high-wealth households are oversampled, aggregate financial wealth captured represents only a fraction of the corresponding totals in national accounts (Flores and Gutiérrez, 2021). Consistent with this shortfall, the survey yields a top 1% wealth share of about 15% throughout most of the period. This appears implausibly low when compared to estimates of income concentration, placing the top 1% national income share above 22% between the same period (Flores et al., 2020; Fairfield and De Luis, 2016; Lopez et al., 2016).¹⁶

Second, the survey substantially understates aggregate household wealth relative to national accounts (Figure B.2a versus Fig. 1(a)). Moreover, while one would expect high-wealth households should hold a larger share of their portfolios in financial assets, the survey only weakly captures this gradient (Figure B.3), suggesting large under-reporting of financial assets at the top. Appendix B.1 discusses these issues in more detail.

Third, the fraction of respondents reporting a positive pension balance is always far below administrative benchmarks. For instance, in 2014 close to 72% of adults held pension funds, yet only 14.3% of survey

¹⁴ Following Piketty and Zucman (2014), we include NPISH because the boundary between individual wealth and private foundations is not clearly defined.

¹⁵ As emphasized by Walzl (2022), the valuation of unlisted shares remains uncertain even in national accounts because market values are not always directly observable. In Chile, medium and small size unlisted shares are typically reported at book value in the Financial Accounts. We propose an alternative valuation in Appendix A.2, using price-to-income ratios from listed firms to adjust the value of unlisted shares based on aggregate administrative sources. These estimates are largely experimental: given the lack of direct information for Chile, market value of unlisted shares could plausibly be above or below book values, so the direction of the bias is unclear. Under this adjustment, the value of financial assets rises by approximately 65–100%, depending on the year.

¹⁶ For comparison, Saez and Zucman (2016) report that in the United States the top 1% holds about 16% of income but over 40% of wealth, while the corresponding figures are 20% and 45% in Colombia (Londoño-Vélez and Ávila-Mahecha, 2021), and 19% and 55% in South Africa (Chatterjee et al., 2022).

respondents reported any pension wealth (Table B.1). Pension balances are collected from a single respondent in the household, referring to their individual accounts rather than household aggregates.¹⁷ The survey can thus hardly support reliable estimates of the pension wealth distribution.

Relative to other Latin American wealth surveys, the Chilean version stands out for its panel structure and partial oversampling of higher-wealth households. However, similar to Mexico's ENFIH, Colombia's IEFIC, and Uruguay's EFHU, it relies on a single respondent and underreports key assets, most notably private pensions (Gandelman and Lübker, 2024). In contrast, benchmark surveys in advanced economies explicitly address these issues. The U.S. Survey of Consumer Finances combines a representative sample with a targeted oversample of wealthy households based on tax data, while the European HFCS and the EU-SILC wealth module rely on top oversampling and multiple imputation to improve coverage of the upper tail (Bricker et al., 2017; Vermeulen, 2018). The Chilean version lacks both administrative linkages and effective top-tail oversampling, resulting in systematic underrepresentation of high-wealth households and pension assets. These limitations motivate our methodological approach, described below.

2.3.2. Forbes data

To correct for upper-tail undercoverage, we use the annual Forbes World Billionaires List (USD 1 billion+) and retain Chilean nationals. This approach has important limitations: Forbes' wealth estimates reflect heterogeneous sources, shifting collection methods, and incomplete coverage (Bach et al., 2019), and exchange-rate movements can distort comparisons over time (Waltl and Chakraborty, 2022). National rich lists could mitigate some of these issues by offering broader coverage and avoiding exchange-rate noise, but for Chile they are available only for a shorter period (2011–2017). Despite these caveats, we use Forbes as an external benchmark, matching each end-of-year survey wave to the following year's list and converting reported dollar wealth into Chilean pesos using Forbes' market closing dates and exchange rates.

2.3.3. Administrative data on pension funds

Since 2007, the Superintendency of Pensions has published a monthly random sample of 5% of active affiliates. It covers roughly 200,000–500,000 individuals per year and provides a monthly panel, yielding about 2–5 million observations annually. The data include accumulated pension balances and key covariates, such as gender, age, nationality, months of formal employment, and reported labor income (the base for the mandatory 10% contribution).

We use these records to impute pension balances from individual characteristics, estimating the imputation model separately by year. Pension wealth is measured as the accumulated balance at year-end, and relies on average monthly labor income to mitigate the influence of unemployment spells on the estimation. While the administrative data lack the richer household detail available in the survey, they offer stable coverage over time and allow us to impute pension wealth consistently for all survey waves.

3. Methodology

Our procedure to create a micro-macro consistent wealth distribution involves four main steps:

1. Pension wealth is imputed at the individual level using four alternative approaches, each evaluated against a benchmark to identify the best-performing method. The resulting predicted values are then aggregated to the household level to construct the distribution of household pension wealth.

2. Household wealth totals are proportionally upscaled to match national aggregates, with separate adjustments for real assets, financial assets (excluding pension assets), pension assets, and liabilities.
3. Data from the rich list are integrated into the survey by fitting a Pareto model to estimate the upper tail, generating synthetic households between the richest household in the survey and the poorest billionaire in the rich list.
4. The imputed, upscaled, and top-corrected datasets are consolidated at the component level. Since the top correction can alter aggregate proportions, we iterate steps 2 and 3 until the resulting totals converge to macroeconomic aggregates. Then, we impute cash using a cell based imputation approach.

These steps are discussed in more detail in the following subsections.

3.1. The imputation of pension funds

We impute pension balances for all survey household members before aggregating wealth to the household level. Our method follows a two-sample, two-stage machine-learning approach, in which an auxiliary dataset is used to train a predictive model that imputes missing variables in the target survey. Following Britto et al. (2025), we implement an intensive-margin strategy, assuming that most individuals accumulate at least some pension wealth over the life cycle through formal-employment contributions.¹⁸

Importantly, this approach may assign positive pension wealth to some individuals who hold zero balances. If this occurs systematically at the lower end of the distribution, it could mechanically compress the distribution and bias inequality downward. In our setting, the main reason this is unlikely is that non-affiliation is concentrated in specific age groups rather than reflecting informality. First, coverage is around 95% for working ages 25 to 55, while non-affiliation is concentrated above age 60 and, to a lesser extent, among the youngest cohort (see Table B.2). Second, our estimation sample includes at least some individuals with zero or near-zero pension wealth. Since age is a key predictor, predicted values for young individuals are typically very small, and this also constrains imputations among older cohorts where non-affiliation is most prevalent. Any remaining concerns are therefore best viewed as concentrated among poorer older households with weak contribution histories.¹⁹ To assess the quantitative relevance of this channel, in Figures D.6 and D.7, we report a robustness check in which we avoid imputing pension assets above retirement age, our main results remain unchanged.

In our setting, the predictors include age, gender, labor income, nationality, and region of residence. We train four families of prediction models, tuning their hyperparameters to minimize cross-validated prediction error: (i) Relaxed LASSO, (ii) Relaxed LASSO with nested cross-validation, (iii) Random Forests, and (iv) a cell-based imputation procedure similar to that used by Garbinti et al. (2021) and Martínez-Toledano (2023) to impute housing wealth. For each method, we estimate several specifications combining different subsets of covariates. Model selection within each family is based on cross-validation in the administrative sample, choosing them based on minimizing the root mean squared error (RMSE) and maximizing the coefficient of determination (R^2). Since the household survey contains partial information on individual pension balances, we use these self-reported values as an out-of-sample validation benchmark to identify the most

¹⁸ We exclude from the imputation individuals who report no affiliation or affiliation with other systems, such as the pre-AFP system and the pension systems of the police and armed forces.

¹⁹ Delayed labor-market entry due to prolonged education is more common among individuals from middle- and upper-wealth households, implying that the absence of pension balances does not necessarily correspond to low household wealth.

¹⁷ While most survey questions refer to all household members, the data is published with hot-deck and MICE (multiple imputation by chained equations) imputations at the household level for most assets and liabilities, but only at the household-head level for pension funds.

Table 1
Predictive accuracy of imputation methods.

	Relaxed Lasso		Multiple Imputation (Means)	Random Forests	Relaxed Lasso		Multiple Imputation (Means)	Random Forests
	K-fold CV	Nested CV			K-fold CV	Nested CV		
	Out of sample RMSE				Sample RMSE			
2007	15.42	14.98	13.52	13.78	5.11	7.47	4.99	4.86
2011	21.88	22.45	19.91	20.67	6.07	7.29	5.68	5.45
2014	31.83	30.76	27.50	27.89	7.86	9.43	7.96	7.44
2017	42.97	41.70	38.50	38.71	10.01	14.35	9.78	9.50
2021	25.95	25.52	26.35	24.83	12.98	15.64	12.62	12.27
	Out of sample R^2				Sample R^2			
2007	0.09	0.14	0.28	0.27	0.66	0.27	0.67	0.70
2011	0.25	0.20	0.28	0.33	0.67	0.46	0.67	0.70
2014	0.18	0.23	0.38	0.37	0.67	0.51	0.65	0.70
2017	0.04	0.09	0.22	0.22	0.66	0.29	0.67	0.69
2021	0.19	0.19	0.16	0.23	0.60	0.42	0.61	0.64

Notes. This table summarizes the fit of different models (selecting the best according to cross-validation) in the survey subsample of individuals reporting pension funds greater than zero and in the administrative training sample. Regressions in the survey subsample are estimated using survey weights.

accurate imputation method. Further details on the specification of each model and the parameters used for their implementation are provided in Appendix B.2.

Table 1 reports the predictive performance of the best specification for each imputation method. Across years and samples, multiple imputation and random forests achieve lower RMSE and higher R^2 than relaxed LASSO, indicating better predictive performance. The gap between in-sample and out-of-sample results shows that random forests and multiple imputation generalize better, while LASSO tends to overfit.

Overall, the random forest model provides the most accurate and stable predictions across both administrative and survey data. We therefore use random forests as our baseline method to impute individual pension wealth. Predicted values are then aggregated to the household level to construct the distribution of pension wealth.

Using survey responses as validation, the model accurately predicts accumulated pension funds across most of the distribution, but tends to underestimate balances in the top decile, a limitation shared by all our prediction models (figures B.6 and B.7). One possible explanation is that either the survey or the administrative dataset provides limited representation of individuals at the very top of the wealth distribution, implying that the imputation is performed in a region with weak or no common support between the administrative and survey data. Another explanation is that the prediction variables used do not sufficiently distinguish individuals with very high pension wealth from the rest of the population and therefore end up underestimating those values.

These concerns are amplified in the top-correction step, where we must assign pension wealth to the synthetic households generated by the Pareto interpolation above the survey's support—that is, households that are, by construction, richer than the wealthiest observed in the survey. Because there is no direct evidence on pension portfolio composition in this region of the distribution, we rely on extrapolation as a partial remedy. Specifically, we assume that the pension asset share of these households is similar to that observed at the very top of the survey distribution. In practice, we set this share equal to the pension asset share (relative to total household assets) observed among households in the top 1 percent prior to any correction.

3.2. Upscaling to national aggregates

To compute trends in wealth inequality that are consistent with the macroeconomic wealth aggregates, we upscale wealth totals in the pension-imputed survey to match national aggregates. Adjustment factors are constructed separately for real estate, financial assets excluding cash, and liabilities, and each household's balance sheet is rescaled

accordingly.²⁰ This procedure follows Vermeulen (2018), Bach et al. (2019), Wodrich (2020), and Waihl and Chakraborty (2022), which apply similar adjustments to these broad categories. Importantly, we separate financial assets from pension funds. This distinction is crucial in countries with private pension systems, as the dynamics of mandatory pension wealth differ from other financial assets: the latter are closely tied to saving capacity and access to capital income, while the former are directly linked to labor income. Thus, poorer households may hold no financial assets but still accumulate pension savings if they participate in the formal labor market.

Because agricultural land is part of national household wealth but is not separately observable in the distributional data, i.e., no rural representation or specific measurement of agricultural assets, we impute it within the housing/real-estate block. This aggregation trades granularity for consistency with macro totals. The direction of any bias is hard to sign since large farms can be extremely valuable, yet rural home-ownership is also relatively high (Table C.3), so the expected distributional impact is unclear a priori. Quantitatively, however, agricultural land accounts for 8.7%–14.8% of real estate wealth, so the induced proportional scaling of real-estate values is small and unlikely to drive our results (Figure A.1).

3.3. Upper tail adjustment with billionaire lists and micro-macro consolidation

We correct for missing wealth at the upper tail of the distribution following the methodology of Vermeulen (2016, 2018). This approach combines the pension-imputed and upscaled household surveys with rich lists that document the wealth of the very wealthy. The two data sources are integrated by fitting a Pareto distribution, which interpolates between the richest households captured in the survey and the poorest individuals appearing in the rich list.

Regarding the number of synthetic households, this quantity is not chosen ex ante and does not reflect an arbitrary assumption about the size of the missing rich. Instead, synthetic households are generated

²⁰ We handle cash separately because it is missing in 2007 and 2011 and, when observed, survey totals fall far short of national accounts. In this setting, a standard uprating would require very large adjustment factors, likely reshuffling households' ranks substantially for an item that is plausibly misreported (e.g., reluctance to disclose cash holdings). Instead, we allocate the national-accounts cash aggregate across the distribution using the survey's within-distribution pattern (ranking households by wealth net of cash), and proxy missing years using the 2021 pattern. Since cash is a very small component of total wealth, this choice is not central to our dynamics and mainly avoids introducing noise. See Appendix B.4 for details.

solely to interpolate between the richest observation in the survey and the poorest individual in the rich list. Their number is endogenously determined by the fitted Pareto tail and by an iterative micro–macro consolidation procedure that ensures consistency with national wealth aggregates by asset class. As a result, the implied size of the missing top is an outcome of the model and the data, rather than an imposed parameter.

Although the formula for the calibration of the Pareto distribution is standard (Appendix B.3), an important methodological choice concerns the determination of the threshold above which observations are assumed to follow the Pareto law. This cutoff must be sufficiently high to ensure that the upper tail conforms to Pareto behavior, yet its implementation is inherently arbitrary. In our benchmark specification, we adopt the common practice of setting it at the beginning of the top 1% of the distribution. To assess the sensitivity of our results, we also report estimates using alternative thresholds ranging from the 80th to the 99.5th percentile. While this choice can influence the measured levels of concentration, it does not materially affect our main findings regarding trends in wealth inequality (Appendix D.16, D.17).

Once the estimated parameters are set, we use them to generate synthetic households that interpolate between the richest households observed in the survey and the poorest individuals captured in the rich list.²¹ We then disaggregate the imputed wealth of these synthetic households and rich-list participants, assuming that their portfolio composition mirrors that of households with comparable ranks—those that belong to the top 1 percent before top correction—in the pension-imputed and upscaled survey (Step-3). Because the upper-tail adjustment increases aggregate wealth, we re-aggregate the top-corrected survey at the component level (Step 2). If residual discrepancies remain, Steps 2–3 are iteratively repeated until convergence is achieved (see Section B.3 for details), ensuring that the uprated and top-corrected household wealth totals by asset category exactly match the corresponding national accounts aggregates. This procedure is applied for all assets and liabilities (excluding cash).

4. The levels of wealth inequality

In this section, we present the estimates of total wealth, wealth inequality, and wealth portfolios for the top 1%, top 10%, middle 50–90 and bottom 50%. We compare wealth totals, and levels with the US and selected Western European countries.

4.1. The wealth-income ratio

Private wealth in Chile expanded remarkably over the past decade and a half. As shown in Fig. 2, the ratio of private wealth to national income rose from roughly 3.2 to 4 times between 2007 and 2021, driven primarily by the rapid appreciation of housing assets—especially after 2014. The acceleration in real estate prices can be partly attributed to the introduction of a 19% value-added tax (VAT) on new dwellings in 2016 and its subsequent pass-through to housing prices (Lozano and Idrovo, 2024).

This upward trend reversed toward the end of the period. Between 2020 and 2021, the wealth-to-income ratio declined from 4.4 to 4.0, reflecting a sharp contraction in pension assets. The stock of pension wealth fell by approximately 5% in 2020 and an additional 18% in 2021, following the extraordinary pension withdrawals authorized as part of the government's emergency response to the COVID-19 crisis.

Housing and pension assets together account for the majority of household wealth in Chile. Housing represents roughly half of total private wealth, while pensions comprise between 26 and 32 percent. In 2021, these two components jointly accounted for about 79 percent of

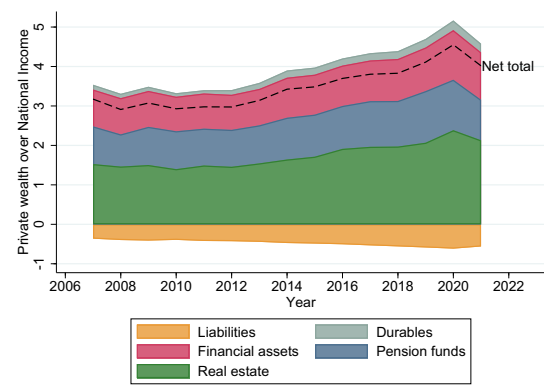


Fig. 2. The evolution of Chilean private net wealth (2007–2021). **Notes.** Elaborated by the authors using data from financial accounts (Central Bank), the housing price index (Central Bank and the National Tax Agency), and Flores et al. (2018). The market value of total wealth, net of liabilities, is approximately four times the National Income in 2021. The “Real Assets” category includes formal housing and agricultural land, the “Financial Assets” category comprises stocks, cash and other financial assets.

the national wealth stock. Financial assets, which include business equity and deposits, account for about 30 percent of total wealth, while durable goods represent around 4 percent. Finally, household liabilities amount to roughly 14 percent of total private wealth.

Alternative estimates presented in Appendix D confirm the robustness of these results. To account for potential undervaluation of unlisted non-financial business assets, we first follow Albers et al. (2024) in computing price-to-earnings ratios from the Chilean stock market and applying them to aggregate tax data on firm earnings. This adjustment raises the wealth-to-income ratio to 4.3 in 2021. We also consider an alternative measure of housing wealth based on the experimental series produced by the Central Bank of Chile.²² Both adjustments yield broadly consistent patterns, as shown in Figures D.8 and D.10. Although the level of aggregate wealth varies slightly across specifications, the overall upward trajectory, its subsequent moderation, and its broad composition remain unchanged.

4.2. Top wealth inequality

We examine top wealth shares, that is, the share of total wealth held by the top 1% and the top 10% wealthiest households. Our estimated series are showcased in Fig. 3.

Fig. 3 shows that the wealth share of the top 10 percent declined from 64.1 percent to 59.7 percent between 2007 and 2021, while the share of the top 1 percent fell from 35.5 percent to 31.6 percent. These levels place Chile close to the United States in terms of wealth concentration, with comparable top 1 percent shares in recent years (Smith et al., 2023). For comparison, we also report estimates that exclude pension assets from total wealth. Excluding pensions increases the top 10 percent's share by 9.6 to 14 percentage points and the top 1 percent's share by 7.4 to 12 points, underscoring the strong equalizing role of mandatory private pensions.

This equalizing effect stems from the institutional design of Chile's pension system: mandatory contributions are linked to labor income, which is less concentrated than capital income. Consequently, pension wealth reduces overall wealth inequality even though it is privately held. This mechanism parallels evidence from the United States, where Saez and Zucman (2016) attribute much of the mid-twentieth-century decline in wealth inequality to the expansion of private pension plans.

²¹ Walti and Chakraborty (2022) propose alternative approaches, such as quantile regression, which may exhibit favorable statistical properties.

²² These series are based on a methodology similar to ours but are available only from 2012 onward. We therefore assume that their pre-2012 trend mirrors that of our estimates.

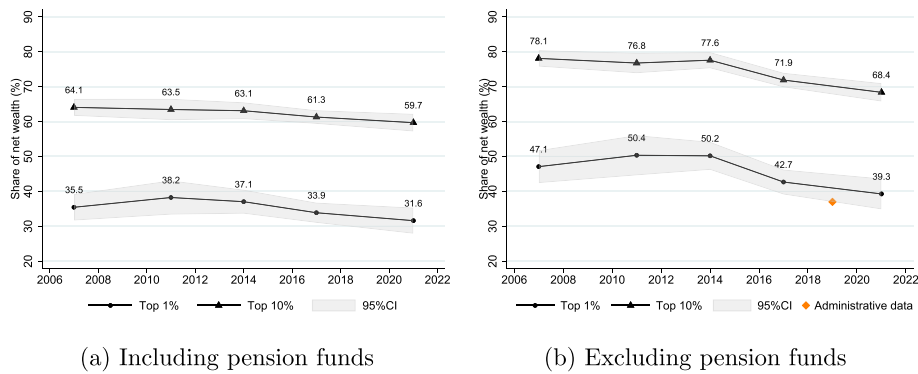


Fig. 3. Adjusted wealth shares, 2007–2021. **Notes.** Own elaboration based on the country's household survey, administrative pension fund records, and Forbes data. In panel (a), pension funds are imputed using Random Forest regression. Standard errors are computed using the analytical methodology proposed by Jann (2016). Pension funds are imputed for all adults, regardless of whether they report pension funds. Cash is aggregated after the top correction as a share of percentile and is imputed for 2007 and 2011 based on the distribution of 2021. Administrative data come from Carranza et al. (2025).

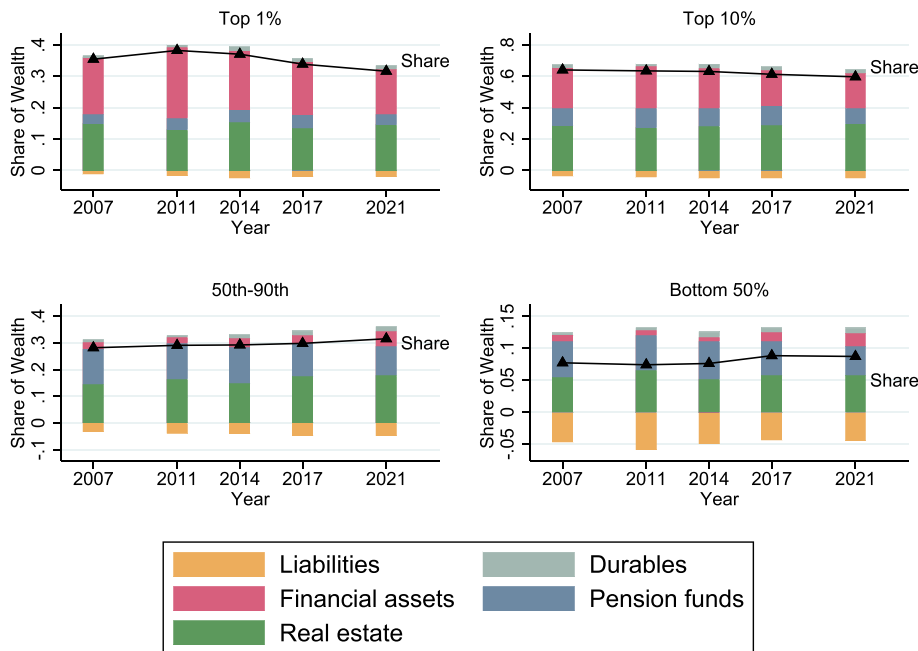


Fig. 4. Portfolios of wealth groups over time. **Notes.** The sum of the heights of all assets minus the height of liabilities, corresponds to each group's wealth share.

We benchmark our estimates against those obtained from administrative tax records for the year 2020. As shown in Fig. 3(b), the top 1% share is close to 37%, 2.3 percentage points below our corresponding estimate for 2021. This comparison indicates that our approach—combining multiple methods with household survey information—closely tracks the distributional patterns derived from pure tax-record-based estimates.²³

4.3. Wealth portfolios by distributional group

Fig. 4 depicts the asset portfolio composition of the top 1 percent, top 10 percent, middle 40 percent (percentiles 50–90), and bottom 50 percent of the wealth distribution. Across all groups, pensions represent a substantial share of total assets: both the bottom 50 percent and the middle 50–90 percent hold more than 40 percent of their assets in

pension funds. Real estate constitutes the main asset class throughout the distribution, accounting for more than half of total assets for all groups. In contrast, financial assets are highly concentrated at the top: they represent about one-third of the portfolio of the top 10 percent and more than 45 percent for the top 1 percent.

These patterns highlight that housing and pensions are the key wealth components for the bottom 90 percent of the population. The bottom 50 percent holds between 7 and 9 percent of aggregate wealth over the period, almost entirely through housing and pension assets; this share is consistent with the high homeownership rates observed across the household income distribution displayed in Table C.3. The middle 40 percent additionally holds deposits and durable goods. At the top of the distribution, the concentration of business and financial assets explains the persistence of wealth inequality, consistent with the extreme levels of income concentration documented for Chile—where the top 1 percent captures around 25 percent of total income, largely through capital income (Flores et al., 2020; De Rosa et al., 2024).

Because the survey is urban-only, our national estimates assume that rural households have similar wealth and portfolio composition to urban households at comparable wealth ranks. To assess how strong

²³ Carranza et al. (2023) note that these figures are slightly underestimated because the denominator includes assets held by the non-profit sector, which are not captured in the numerator. This discrepancy can explain the divergence from our estimated values.

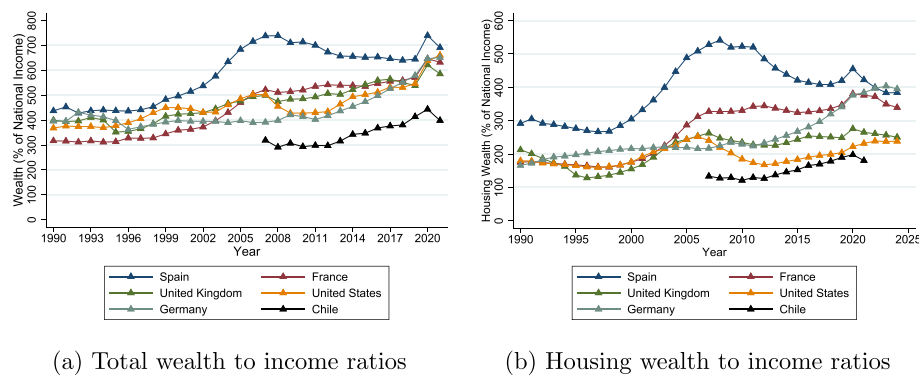


Fig. 5. Wealth to income ratio in an international context. **Notes.** Private wealth–income ratios, except for Chile, come from WID.world and draw on the works of Saez and Zucman (2016) (USA), Garbinti et al. (2018) (France), Alvarado et al. (2018) (UK), Martínez-Toledano (2023) (Spain), and Albers et al. (2024) (Germany). The ratios for Chile are constructed by the authors. Housing wealth does not include agricultural wealth.

this assumption is, we use the nationally representative CASEN survey, which measures income rather than wealth. Tables C.3 and C.4 show higher homeownership and lower labor-market formality at comparable income levels in rural areas, implying more housing wealth and less pension accumulation throughout. Since the rural population is small and poorer on average, its inclusion would likely change wealth shares only modestly. In an extreme scenario where 75% of rural households have zero wealth, the bottom-50% share declines by only 1.5 percentage points.

4.4. Chilean wealth in an international context

Fig. 5 compares Chile's wealth-to-income ratio with those of several developed economies. As shown in panel (a), Chile's ratio increased from roughly 3 to 4 over the study period, whereas in advanced economies it rose from about 4 to 7. We argue that these differences primarily reflect structural factors—particularly the behavior of housing prices.²⁴ Prior to the 2000 housing boom in France, the wealth-to-income ratio was about 3.5; in Germany, it hovered around 4 before the sharp post-2010 housing price surge; and in Spain, it was approximately 4.5 before the late-1990s boom. After these housing booms, the ratio exceeded 7 in Spain, rose to 5 in France, and surpassed 6.5 in Germany. Panel (b) corroborates this pattern by showing that the private housing wealth-to-income ratio increased from nearly 3 to more than 5 in Spain, from 1.7 to over 3 in France, and from below 2.4 to more than 4 in Germany following their respective housing booms.

By contrast, Chile exhibits the lowest housing wealth-to-income ratio among the countries considered, even after the post-2014 increase in housing prices, which rose from 1.5 to 2. This rise, though significant, was far less pronounced than in France, Germany, or Spain. The absence of an extreme housing boom helps explain Chile's comparatively lower aggregate wealth-to-income ratio.²⁵ When compared with the United States and the United Kingdom, Chile's 2020 ratio is broadly similar to their respective levels around 2000. The remaining gap with the United States reflects differences in financial-asset holdings, while in the United Kingdom it arises from both housing and financial assets.

Fig. 6 situates Chile's wealth distribution in an international context. As shown, Chile records the highest share of wealth held by the bottom 50%, reaching nearly 9% in 2021. In contrast, the middle 40

percent in Chile holds a smaller share than its counterparts in European countries but a larger one than in the United States, particularly in recent years. This suggests that Chile's middle class is relatively smaller than in Western Europe but somewhat larger than in the U.S. Over the same period, the share of the top 10% declined by about four percentage points, placing Chile close to European levels, roughly 11 points below the U.S. in 2021. However, the top 1% share remains comparable to that of the United States and higher than in Europe, although it shows a sharp decline between 2014 and 2021.

These patterns can be understood in the context of Chile's privatized welfare model, which promotes home ownership among the bottom 50% and relies on a mandatory private pension system. This structure encourages widespread asset ownership among low- and middle-wealth households while discouraging additional financial saving, particularly within the middle 40 percent. As a result, capital income and business asset ownership are extremely concentrated at the top. Wealth inequality in Chile is therefore more moderate among the bottom 90%, yet extreme within the top decile, resembling the U.S. more than Europe in that aspect.

Regarding wealth components, the portfolio composition of the bottom 50% in Chile differs markedly from that in Germany, where this group primarily holds deposits and other financial assets, alongside some housing (Albers et al., 2024). A similar pattern is found in France, where the bottom half of the distribution holds mostly deposits and housing (Garbinti et al., 2018). In contrast, Chile more closely resembles Spain, where wealth among the bottom 50% is concentrated in housing assets (Martínez-Toledano, 2023). This can be explained by the accessibility of home ownership across the distribution generated by the ownership oriented government housing policies. In addition, Figure C.5 separates financial assets between cash and other financial assets. We can see that cash only increases after the pension fund withdrawals, indicating that households may adjust intertemporal savings after the 2020–2021 pension withdrawals.

4.5. Robustness checks

We conduct a series of robustness checks to assess the sensitivity of our distributional results. Across specifications, wealth concentration remains highly stable in both levels and trends: between 2007 and 2021, the top 1% share lies between 31–40% and the top 10% between 59–66%.

Imputing pension wealth only for individuals with zero, missing, or non-reported balances—while preserving self-reported positive holdings—yields top shares of 32.1–35.2% (top 1%) and 61.2–64.1% (top 10%), indicating that full pension imputation does not compress inequality (Figure D.5). Replacing housing wealth with the Central Bank's experimental valuation slightly raises total wealth but leaves the top 1% share within one percentage point of baseline, while redefining pension

²⁴ It is reasonable to assume that part of these differences may stem from measurement error in the Chilean wealth stock. To assess this possibility, we estimate an alternative series of wealth stocks, which raises the wealth-to-income ratio by between 0.2 and 0.6 points.

²⁵ This pattern aligns with differences in population density: Chile has roughly 26 people per km², compared with 96 in Spain, 124 in France, and 240 in Germany (World Bank, 2022).

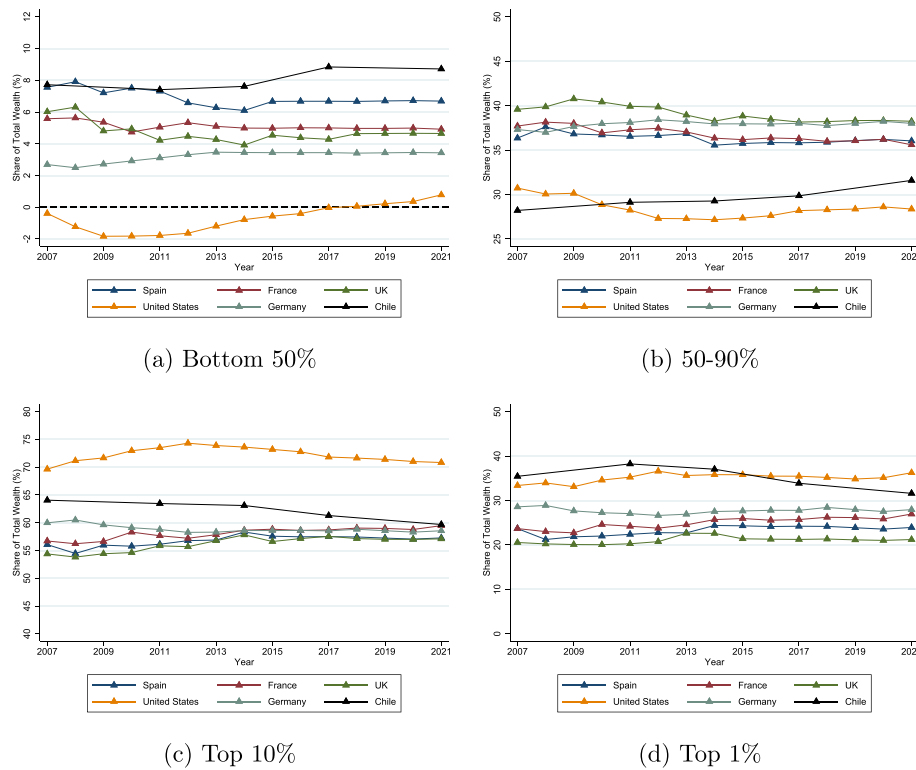


Fig. 6. Shares of wealth by group, international comparison. **Notes.** Wealth shares by quantile, except for Chile, are from [WID.world](#) and draw on the works of [Saez and Zucman \(2016\)](#) (USA), [Garbinti et al. \(2018\)](#) (France), [Alvaredo et al. \(2018\)](#) (UK), [Martínez-Toledano \(2023\)](#) (Spain), and [Albers et al. \(2024\)](#) (Germany). The ratios for Chile correspond to those shown in the baseline of our estimates.

assets to include only mandatory balances produces only minor changes (Figures D.9 and D.14).

Adjusting business assets following [Albers et al. \(2024\)](#) increases non-financial corporate equity by factors of 1.29–2.08. This raises the top 1% share from 31.9% to 35.7% and the top 10% from 60.1% to 64.7%, without affecting trends (Figure D.11).

Across imputation methods—Relaxed LASSO (K-fold and nested), Random Forests, and cell-based means—the top 1% share ranges from 31.3% (2021) to 39.3% (2014), and the top 10% from 57.6% to 65.7% (Figure D.15). Without uprating and using survey totals only, the top 1% share lies between 32–36% (40% in 2014) and the top 10% between 60–62% (64.5% in 2014), showing that macro adjustments mainly affect levels rather than distribution (Figure D.18).

Reversing the order of adjustments—applying Pareto interpolation before macro upscaling—yields similar patterns: the top 1% share peaks at 45.3% in 2011 and declines from 37.7% to 33.8%, while the top 10% peaks at 67.5% and falls from 65.8% to 60.8% (Figure D.20).

Omitting cash imputation has negligible effects, with the top 1% share falling from 36.4% to 33.5% and the top 10% from 65.1% to 61.5%. Excluding durable goods raises the top 1% share by about one percentage point (Figures D.21 and D.22).

5. Wealth inequality dynamics: housing gains and pension losses

Because our estimation of wealth inequality involves several steps and assumptions, we can only hypothesize that recent pension and tax reforms influenced its evolution in Chile. To assess the plausibility of this conjecture, we quantify the potential effects of the 2014 VAT reform on new dwellings, and the 2020–2021 pension withdrawals by conducting a counterfactual analysis based on our decomposition of assets and totals.

First, notice that wealth share can be expressed as a function of the aggregate share of each asset and the share of that asset held by the group, following [Hempel \(2022\)](#) and [Advani et al. \(2025\)](#). Indeed, let

$f \in F$ denote a fractile of the wealth distribution, and let W^f be the sum of individual wealth across all households in that fractile. The share of total wealth sh^f accruing to fractile f is given by:

$$sh^f = \frac{W^f}{W}, \quad (2)$$

where W denotes aggregate wealth. Note that W^f can also be expressed as the sum of all asset components W_j^f belong to that fractile. If is the sum of all asset components W_j , then:

$$sh^f = \sum_j \frac{W_j^f}{W_j} \cdot \frac{W_j}{W} = \sum_j \psi_j^f \omega_j, \quad (3)$$

where ω_j represents the aggregate share of asset j in total wealth, and ψ_j^f is the share of asset j held by fractile f .²⁶

Thus, our decomposition proceeds in two steps. First, we construct a counterfactual share of asset j held by group f , $\tilde{\psi}_j^f$. We use 2014 as the base year and reweight its observations to match the covariate distribution of 2017 or 2021, thereby holding the 2014 structure fixed while adjusting for compositional change. We implement the [DiNardo et al. \(1996\)](#) (DFL) reweighting method.²⁷ Second, we construct

²⁶ For example, if j represents real estate, then $\omega_j = 0.5$ indicates that real estate constitutes 50% of total assets in the economy. If f corresponds to the top 10%, $\psi_j^f = 0.5$ would imply that this group owns 50% of all real estate. In this sense, the wealth share of a given group can be viewed as a weighted average of asset shares, where the weights reflect the aggregate composition of the asset portfolio. Figure C.3 shows ψ_j^f for all of our analyzed years and fractiles. Appendix C discusses the levels and trends of ω and ψ between 2007 and 2021.

²⁷ This approach has been used in the wealth-inequality literature; see [Davies et al. \(2017\)](#) and [Albers et al. \(2024\)](#). Appendix B.5 describes our methodology in more detail.

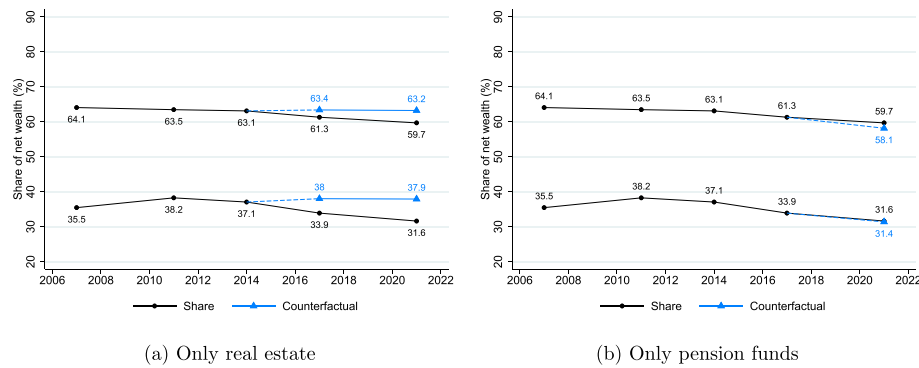


Fig. 7. Counterfactuals of top shares. **Notes.** Own elaboration based on the baseline estimations from Fig. 3 and the methodology of DiNardo et al. (1996). Panel (a) reports the counterfactual scenario in which real estate wealth follows a trajectory similar to the pre-VAT reform period, while Panel (b) reports the counterfactual in which pension funds evolve as they did before the COVID-19 withdrawals.

counterfactual asset totals by computing the share of each asset j in aggregate wealth, $\tilde{\omega}_j$, under two scenarios: (i) real estate wealth grows after 2014 at its pre-2014 rate, and (ii) pension wealth grows after 2019 at its pre-2019 rate. Using these projected aggregates and the counterfactual within-asset allocations from step one, we obtain counterfactual group wealth shares $sh^{f,c}$ as:

$$sh^{f,c} = \sum_j \tilde{\psi}_j^f \cdot \tilde{\omega}_j. \quad (4)$$

This decomposition allows us to distinguish compositional changes in wealth within each group from macroeconomic changes in the aggregate wealth portfolio.

From Fig. 7, we can see that when we assume real estate continued to grow at its pre-2014 rate, the top shares rise: both the top 1% and the top 10% increase relative to the baseline.²⁸ In particular, the top 1% share increases by 6.3 percentage points, and the top 10% share increases by 3.5 points. For pensions, maintaining the pre-2019 growth path implies a larger decline in the top 10% share than we estimate in the baseline. More broadly, as shown in Appendix D.2, our central conjecture—that the increase in housing prices, greatly explained by the VAT reform, lowered wealth inequality, whereas the 2020–2021 pension withdrawals raised it—proves robust to alternative specifications and to variations in each step of our wealth-estimation procedure.

To further understand this result, we compute an additional decomposition:

$$sh^{f,c} = \sum_j \tilde{\psi}_j^f \tilde{\omega}_j = \underbrace{\sum_j (\tilde{\omega}_j - \omega_j) \psi_j^f}_{\text{Change in } \omega} + \underbrace{\sum_j (\tilde{\psi}_j^f - \psi_j^f) \tilde{\omega}_j}_{\text{Change in } \psi} + \underbrace{\sum_j \psi_j^f \omega_j}_{\text{Effective share}}. \quad (5)$$

Eq. (5) expresses the counterfactual share of group f , denoted $sh^{f,c}$, as the sum of three components: the change explained by the evolution of aggregate asset composition (ω_j), the change driven by the within-asset redistribution across groups (ψ_j^f), and the original effective share sh^f . The first component captures how shifts in the relative weight of each asset class affect the overall wealth distribution. The second component isolates distributional changes arising from heterogeneous variation in the value of asset j across the wealth distribution.

To illustrate, consider a housing boom. The first effect operates mechanically: because housing is more evenly distributed than financial assets, an increase in the aggregate importance of housing tends to reduce overall concentration. The second effect reflects differential price appreciation across dwellings. Not all properties increase in value at the

Table 2
Decomposition of counterfactuals.

Year	Quantile	Effective share	Counterfactual share	Change in ω	Change in ψ
Housing counterfactual					
2017	Top 10%	61.30%	63.38%	0.23%	1.86%
2017	Top 1%	33.89%	38.01%	0.31%	3.82%
2021	Top 10%	59.69%	63.22%	0.12%	3.41%
2021	Top 1%	31.62%	37.91%	0.14%	6.15%
Pension funds counterfactual					
2021	Top 10%	59.69%	58.15%	−1.47%	−0.07%
2021	Top 1%	31.62%	31.38%	−1.30%	1.06%

Notes. Columns 5 and 6 represent the terms of Eq. (5).

same rate, and appreciation may be stronger either at the bottom or at the top of the distribution. This heterogeneity generates distributional impacts distinct from the mechanical composition effect and can either amplify or mitigate overall wealth inequality.

Table 2 reports the decomposition for real estate and pension assets. For real estate, the decline in inequality is explained primarily by a relative redistribution of real estate wealth toward the bottom of the distribution. This redistributive component grows between 2017 and 2021, suggesting that real estate appreciation during this period was stronger for dwellings held by the bottom 90% than for those owned by the upper tail. For pensions, the pattern is reversed. Pension withdrawals generate an increase in inequality, driven almost entirely by the mechanical composition effect: as pension balances, an asset class that is comparatively evenly distributed, shrink relative to total wealth, overall wealth concentration mechanically rises. The within-asset redistribution component plays a much smaller role, indicating that the withdrawals reduced pension balances broadly across the distribution.²⁹

To complement these results, we replicate the full set of robustness exercises for the counterfactual analysis under alternative specifications (Section D.2). Holding constant different macro aggregates, such as freezing real estate at its pre-policy growth path, yields consistent patterns: the decline in inequality observed after 2016 largely disappears across all specifications. In this scenario, the top 1% share increases by roughly four percentage points, and the top 10% share by about two.

²⁸ Figure C.4 shows the magnitude of using $\tilde{\omega}_j$ as counterfactual for the evolution of total wealth components.

²⁹ Incorporating rural households may reinforce the equalizing impact of the expansion in housing wealth, given higher rural home-ownership rates (Table C.3). Conversely, because rural workers are more often informal and thus accumulate less pension wealth (Table C.4), pension withdrawals may have contributed less to inequality than in an urban-only sample.

Regarding pension withdrawals, our alternative specifications corroborate our conjecture that pension liquidations raise wealth inequality.

6. Conclusion

We document a high wealth concentration in Chile alongside widespread asset ownership at the bottom. The top 1% holds more than one-third of private wealth driven by concentrated financial and business assets. At the same time, the bottom 50% owns 7–9% of total wealth, mainly in housing and mandatory pension accounts. This pattern reflects Chile's ownership-oriented social model: private, fully funded pensions and homeownership subsidies broaden asset ownership among low- and middle-wealth households but do little to dilute concentration at the very top.

Trends over 2007–2021 reflect two opposing forces. Housing appreciation following the VAT reform implemented in 2016 strengthened middle-class balance sheets and reduced inequality. By contrast, the 2020–2021 pension withdrawals lowered pension balances across the distribution and, because pensions are the most evenly held asset, mechanically increased measured inequality relative to a counterfactual without withdrawals. Our DFL counterfactuals confirm that the post-2016 decline in inequality would have been smaller without housing gains and larger in the absence of pension losses.

These findings emerge under severe data constraints. Chile lacks a comprehensive wealth tax and associated microdata, survey coverage is thin at the top, and pension balances are poorly reported. We address these limitations through a micro–macro integration strategy that imputes pension balances from administrative samples, applies Pareto-based top corrections anchored in rich lists, and uprates all aggregates to national accounts. Although our results are robust across specifications, the estimated trends should be interpreted as informed conjectures, given the multiple assumptions and imputations required to construct the wealth distribution.

Importantly, our statistics capture private wealth, not welfare. Cross-country comparisons should therefore be interpreted with care: although bottom 50% wealth shares in Europe may appear similar, European households benefit from more extensive in-kind provision (public pensions, health, and housing) that substitutes for private asset accumulation. Chile's privatized welfare architecture channels household savings toward private assets (pensions and housing) so comparable wealth shares do not translate into comparable economic security.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT to improve language and readability. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at doi:10.1016/j.jpubeco.2026.105665.

Data availability

Data will be made available on request.

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